PassFrame
tuition

Number Sequences & Patterns

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: The nth term of a sequence is 5n − 2. Find the 9th term and state whether 200 is ever a term.

Method & answer: 9th term = 5×9−2 = 43. For 200: 5n−2=200 gives n=40.4, not a whole number, so 200 is never a term.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. Find the nth term rule for the sequence 3, 10, 17, 24, …

2. Find the 15th term of the sequence with nth term 6n − 5.

3. Which is the nth term rule for 9, 16, 23, 30…? A) 7n+2 B) 7n−2 C) 8n+1 D) 6n+3

4. Find the next two terms: −20, −13, −6, __, __.

5. Find the missing term: 6, 15, __, 33, 42.

6. Select ALL the sequences below with a common difference of 11: A) 4,15,26,37 B) 5,16,27,38 C) 3,13,23,33 D) 2,13,24,35 E) 6,17,28,39

7. A sequence has nth term 9n − 6. Find the first three terms.

8. Find the next two terms: 300, 278, 256, __, __.

9. The nth term of a sequence is 12n − 9. Which of these is the 10th term? A) 101 B) 108 C) 111 D) 120

10. Find the missing terms: −8, __, 4, __, 16.

11. Find the next two terms: 4, 4.75, 5.5, 6.25, __, __.

12. Select ALL of these that are terms in the sequence with nth term 8n + 3: A) 11 B) 19 C) 25 D) 27 E) 35

13. Find the nth term rule for the sequence 150, 137, 124, 111, …

14. A sequence has nth term $3n^2 - 5$. Which of these is the 4th term? A) 31 B) 43 C) 48 D) 53

15. Find the next two terms: 3, 8, 18, 33, __.

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. The nth term of a sequence is $n^2 + 3n$. Find the first three terms.

2. Find the next two terms: 3, 9, 27, 81, __, __.

3. A sequence has nth term $4n^2$. Which is the 5th term? A) 20 B) 80 C) 100 D) 400

6. Select ALL the sequences below where each term is multiplied by 4 to get the next: A) 2,8,32,128 B) 3,6,9,12 C) 5,20,80,320 D) 3,12,48,192 E) 7,14,28,56

7. Find the next two terms: 2, 5, 11, 23, __.

8. Find the nth term rule for the sequence 4, 9, 16, 25, 36… and state the 12th term.

11. A sequence has nth term $n^2 - 4$. Select ALL of these that are terms: A) 5 B) 12 C) 21 D) 30 E) 32

12. Find the next two terms: 1600, 800, 400, __, __.

13. The nth term of a sequence is $4n^2 - 7$. Find the 3rd term.

4. Find the missing term: 4, 12, 36, __, 324.

5. A sequence starts at 2000 and each term is half of the one before. Find the 5th term.

9. Which nth term rule generates the sequence 3, 12, 27, 48, 75...? A) $3n^2$ B) n^2+2 C) $6n-3$ D) $3n+9$

10. Find the missing term: 5, 20, 80, __, 1280.

14. Find the missing term: 5, 15, 45, __, 405. A) 90 B) 120 C) 135 D) 150

15. A geometric sequence starts at 5 and each term is multiplied by 3. Find the 6th term.

Standard / Advanced — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. What is the 30th term of the sequence with nth term rule $4n - 3$?

2. The nth term of a sequence is $n^2 - 3$. Which of these is the 10th term? A) 93 B) 97 C) 100 D) 103

3. Two sequences are given by $7n + 2$ and $5n + 16$. For what value of n are their terms equal?

4. A sequence has nth term rule $9n - 4$. Select ALL of these that are terms: A) 14 B) 23 C) 30 D) 41 E) 50

5. A sequence has nth term $200 - 6n$. For which value of n does the term first become negative?

6. A conference hall has 24 seats in row 1, and each row has 6 more seats than the row before, for 20 rows. How many seats does the hall have in total?

7. A sequence of squares uses matchsticks in a row: 1 square uses 4 sticks, and each extra square needs 3 more. Using the nth term rule, how many sticks are needed for a row of 40 squares?

8. A ball dropped from 10m loses 30% of its height on each bounce. Which of these is closest to the height of the 3rd bounce? A) 3.4m B) 4.9m C) 5.9m D) 7.0m

9. Find the nth term rule for the sequence 3, 10, 29, 66, ... (hint: compare to n^3).

10. A sequence has nth term rule $5n^2 + 1$. Select ALL of these that are terms: A) 6 B) 21 C) 46 D) 80 E) 126

11. A theatre reduces its seats each row: row 1 has 60 seats, and each row has 4 fewer seats than the one before. After how many rows does the number of seats first drop to 20 or fewer?

12. A car depreciates by 15% of its value each year. It is bought for £18,000. What is its value after 3 years, to the nearest £100?

13. The nth term of a sequence is $2n^2 - 3n + 1$. Which of these is the 5th term? A) 26 B) 31 C) 36 D) 41

14. A pattern of hexagonal tiles: 1 tile uses 6 edges, and each extra tile added in a row shares one edge, needing 5 more edges. How many edges are needed for a row of 25 tiles?

15. A sequence of numbers is generated by the rule 'triple the previous number and subtract 4', starting at 3. Find the 5th term.